Al-Hayat International School
Aramoun, Mount Lebanon · American High School Program

Photosynthesis: Following the Energy

Biology · Grade 11 AP · Student Inquiry Sheet

Name: _____ **Section:** _____ **Date:** _____

The Bekaa greenhouse. In January, Rami's tomatoes in his greenhouse near Zahleh grow slowly. The days are short and cloudy. He wonders whether to switch on lamps over his plants. By the end of today you will advise him.

Q0. Gut feeling. Will the lamps help? Write your first guess.

The whole story in one line

$$6\,CO_2 + 6\,H_2O \xrightarrow[\text{chlorophyll}]{\text{light energy}} C_6H_{12}O_6 + 6\,O_2$$

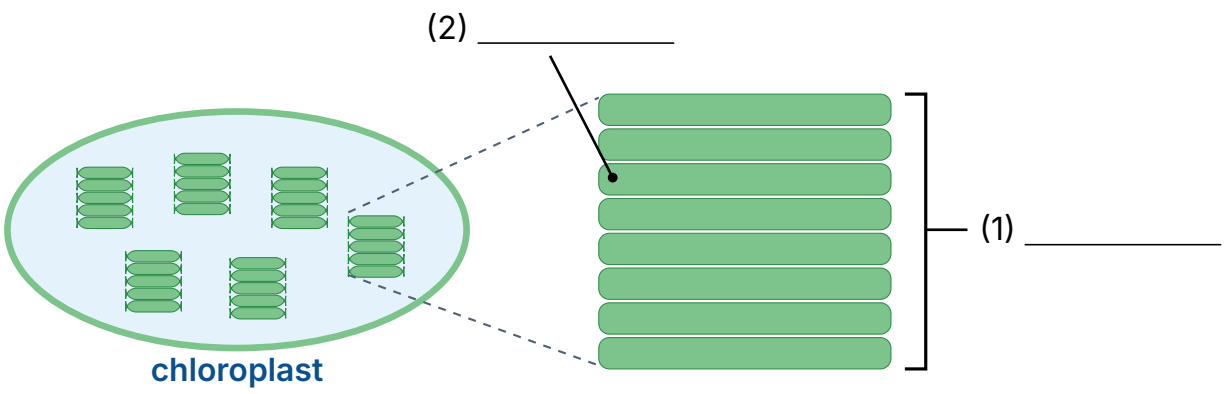
carbon dioxide + water ⟶ glucose + oxygen

A plant does this in two linked stages inside the chloroplast: the **light reaction** and the **dark reaction**.

Part 1 The light reaction

The light-dependent reaction captures the energy of sunlight and stores it in two energy carriers: **ATP** and **NADPH**.

1.1 Location: thylakoids and grana

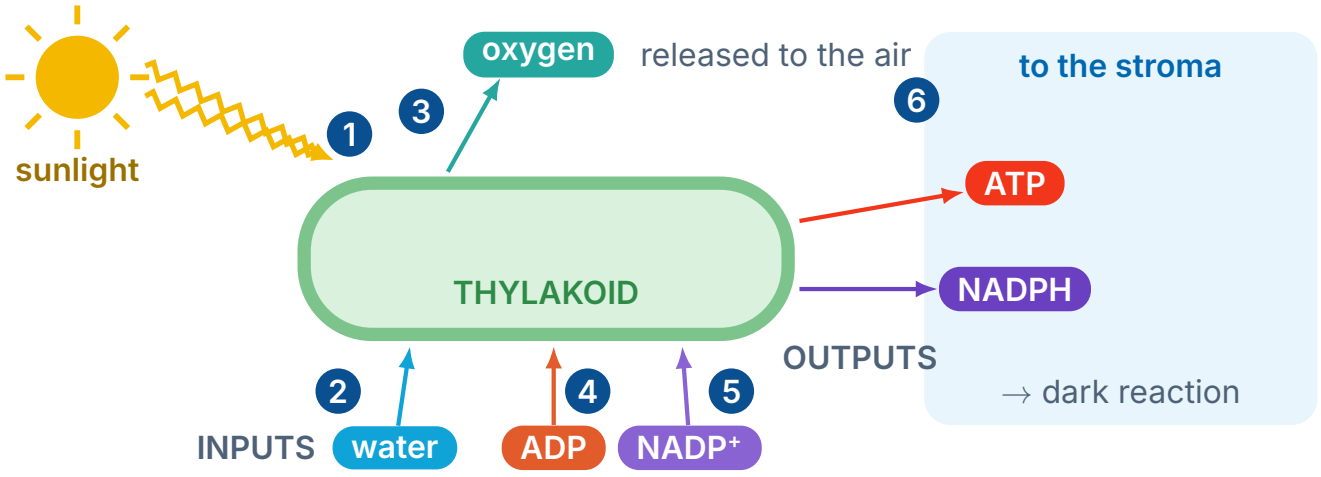


Q1. Label the diagram. Use the words **granum** and **thylakoid**.

Q2. The light reaction takes place in the _____ membranes, which are stacked into _____ .

Q3. Brain teaser: Why do you think a chloroplast has its thylakoids *stacked* into many grana instead of having just one big thylakoid?

1.2 Steps of the light reaction



Q4. Number the steps 1–6 in the order they happen. The numbers on the diagram will help.

- ☐ ADP is charged up to make ATP.
- ☐ Oxygen from the water is released into the air.
- ☐ ATP and NADPH carry the energy to the stroma, for the dark reaction.
- ☐ Chlorophyll in the thylakoid membrane absorbs sunlight.
- ☐ $NADP^+$ is loaded up to make NADPH.
- ☐ The light energy is used to split water.

1.3 The reaction

Q5. Complete the light reaction.

water + _____ + ADP + _____ $\xrightarrow{\text{in thylakoids}}$ _____ + ATP + _____

1.4 Inputs and outputs

Q6. Tick one box for each molecule in the **light reaction**.

	water	sunlight	ADP	$NADP^+$	oxygen	ATP	NADPH
Input	☐	☐	☐	☐	☐	☐	☐
Output	☐	☐	☐	☐	☐	☐	☐

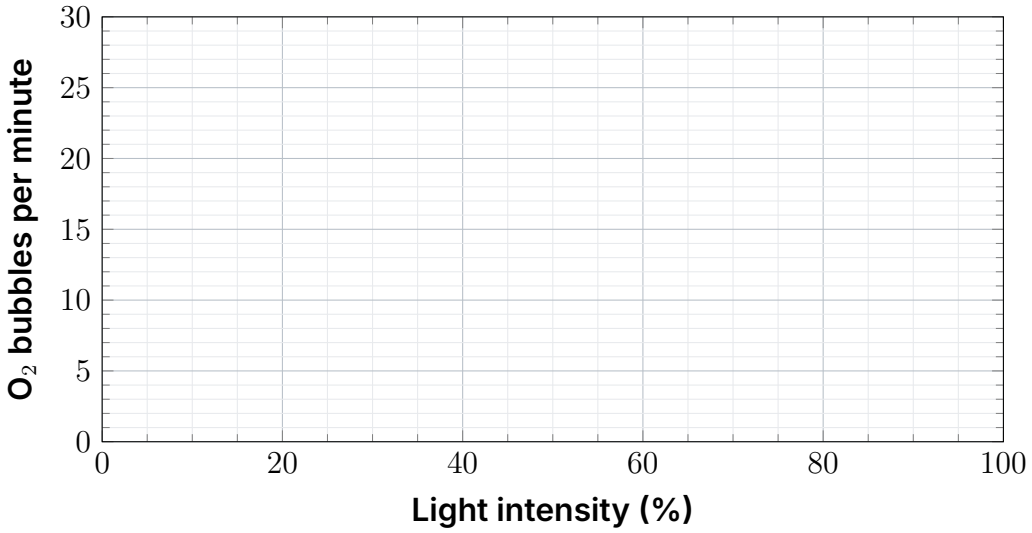
Q7. Brain teaser: Every breath you take contains oxygen made by plants. Look at the inputs: which molecule does that oxygen come from?

1.5 Effect of sunlight intensity

A pondweed is placed under a lamp. The **oxygen bubbles** it gives off show how fast the light reaction runs.

Light intensity (%)	0	20	40	60	80	100
O_2 bubbles per minute	0	15	23	26	28	29

Q8. Plot the data and join the points with a smooth curve.



Q9. Describe what happens to the rate as light intensity increases.

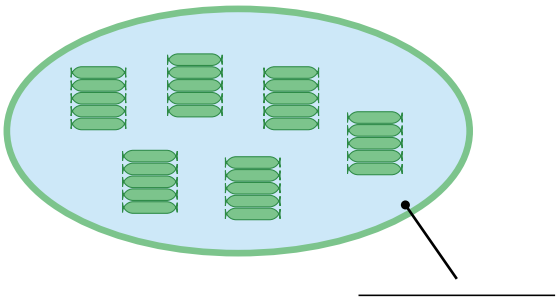
Q10. Brain teaser: Between 80% and 100% light, the plant gets more light but hardly makes more oxygen. Suggest why.

Q11. Real world: Use your graph to explain why Rami's tomatoes grow slowly on short, cloudy January days.

Part 2 The dark reaction

The light-independent reaction (Calvin cycle) uses the energy in **ATP** and **NADPH** to build **glucose** from **carbon dioxide**.

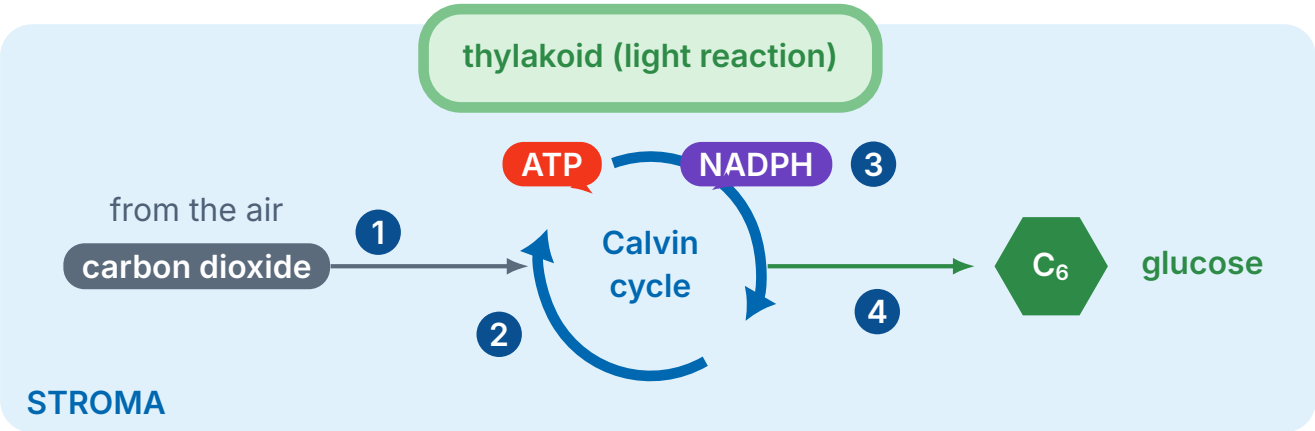
2.1 Location: the stroma



Q12. The dark reaction takes place in the _____ of the chloroplast. Label it on the diagram.

Q13. Why is it useful that the stroma surrounds the thylakoids?

2.2 Steps of the dark reaction



Q14. Number the steps 1–4 in the order they happen.

- ☐ ATP and NADPH supply the energy to build sugar.
- ☐ Glucose is made. The plant uses it or stores it as starch.
- ☐ Carbon dioxide from the air enters the stroma.
- ☐ CO_2 is fixed (joined) into the Calvin cycle.

2.3 The reaction

Q15. Complete the dark reaction.

_____ + ATP + _____ $\xrightarrow{\text{in stroma}}$ _____

2.4 Inputs and outputs

Q16. Tick one box for each molecule in the **dark reaction**.

	carbon dioxide	ATP	NADPH	glucose
Input	☐	☐	☐	☐
Output	☐	☐	☐	☐

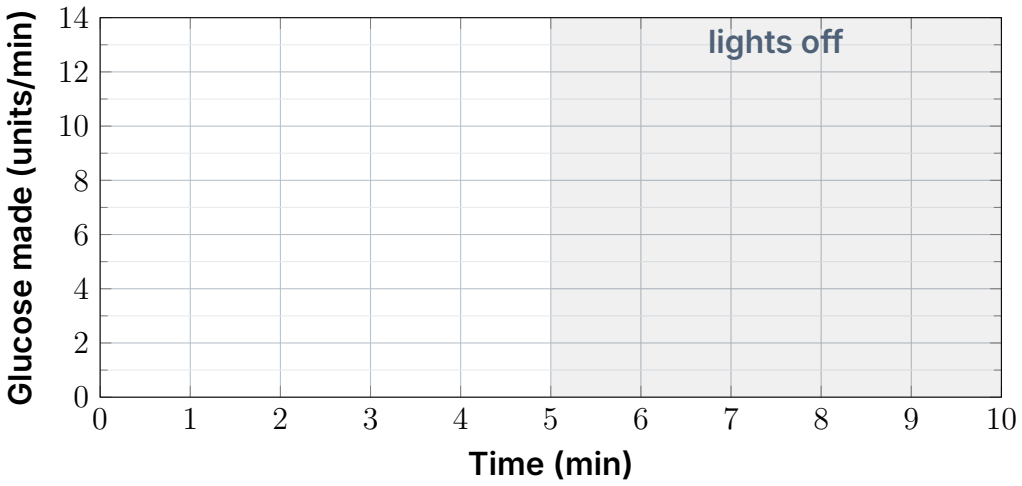
Q17. Brain teaser: The dark reaction does not use light directly. So does it keep making glucose all night long? Explain.

2.5 Effect of available energy

A plant is kept in bright light. At minute 5 the lights are switched off.

Time (min)	0	1	2	3	4	5	6	7	8	9	10
Glucose made (units/min)	12	12	12	12	12	12	6	2	0	0	0

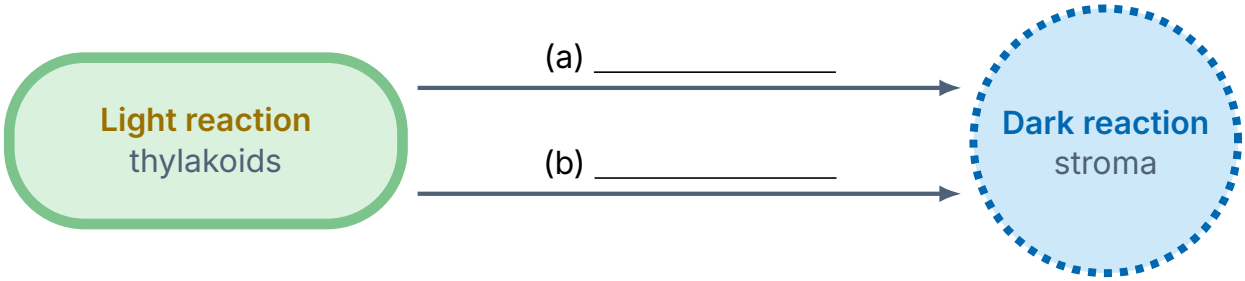
Q18. Plot the data. What happens to glucose production after the light goes off?



Q19. Explain why glucose production stops, even though carbon dioxide is still available.

Q20. Brain teaser: Predict what happens if a plant gets *more* ATP and NADPH than usual. Will glucose production rise forever?

Part 3 Connect the two reactions



Q21. Write the two molecules that carry energy from the light reaction to the dark reaction on the arrows.

Q22. Advise Rami (Claim – Evidence – Reasoning). Should Rami switch on lamps over his tomatoes on cloudy January days? Use your graph from 1.5 as evidence, and explain using both reactions.

Claim:

Evidence:

Reasoning:

Q23. Look back at Q0. Was your gut feeling right? What changed your mind (or confirmed it)?

Exit ticket · 3 – 2 – 1

3 differences between the light and dark reactions

2 molecules that link them _____ _____

1 question you still have